REVISION NOTES

Week 3

Sun 1 – Fri 6 November 2026

15	9 h 33 min	15 / 15
LECTURES	VIDEO	SUMMARISED

COURSES THIS WEEK

MATH 120	Single-Variable Calculus	2 lectures · 1 h 45 min
AI 200	Neural Networks: Zero to Hero	3 lectures · 3 h 30 min
SPCH 100	Storytelling	6 lectures · 1 h 4 min
MATH 130	Probability	2 lectures · 1 h 39 min
MATH 110	Linear Algebra	2 lectures · 1 h 35 min
Total		**15 lectures · 9 h 33 min**

Study days Sun 1 Nov to Fri 6 Nov. Hours are video time; a lecture the plan spreads over several days counts only the days that fall in this week.

WEEK 3 · SUN 1 – FRI 6 NOVEMBER 2026

Contents

LECTURES, IN PLAN ORDER

1	MATH 120	Rate of Change	Sun 1
2	AI 200	Makemore 4 — becoming a backprop ninja (DO NOT SKIP)	Sun 1 – Tue 3
3	SPCH 100	The shapes of stories — fortune drawn over time	Sun 1
4	MATH 130	Conditioning Continued, Law of Total Probability	Mon 2
5	SPCH 100	Building a story live: know the end, start close to it	Mon 2 – Tue 3
6	MATH 110	Multiplication and Inverse Matrices	Tue 3
7	MATH 120	Limits	Wed 4
8	AI 200	Makemore 5 — building a WaveNet	Wed 4 – Thu 5
9	SPCH 100	And, But, Yet, Therefore — structure in one breath	Wed 4
10	SPCH 100	Structure is what they know, and when — the Nemo flashbacks	Wed 4
11	SPCH 100	Write what you know — the feeling under the monster story	Wed 4
12	MATH 130	Monty Hall, Simpson's Paradox	Thu 5
13	SPCH 100	Credibility before vulnerability — five tips for a story in a talk	Thu 5
14	MATH 110	Factorization into A = LU	Fri 6
15	AI 200	Let's build GPT — from scratch, in code	Fri 6

AT THE BACK

What this week established

Answers to the self-checks

Every lecture this week has a summary.

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 1 OF 15

Rate of Change

Sun 1 Nov · 52 min video · Watch on YouTube

THE TAKEAWAY

The derivative is the slope of the tangent line, defined as the limit of secant slopes — and the difference quotient is the formula that makes it computable.

THE ARGUMENT · 6 STEPS

1. **Drawing a tangent by eye is not a definition**
   You can see the right line; a machine cannot. What distinguishes it is not that it touches once — a wiggly curve can be crossed repeatedly. The honest statement is a limit of SECANT lines PQ as Q slides towards P.

2. **Two notations, and then one formula**
   Δx is the horizontal step, Δf the resulting rise, and $\Delta f/\Delta x$ is the secant's slope. Then $f'(x_0) = \lim_{\Delta x \to 0} \dfrac{f(x_0+\Delta x) - f(x_0)}{\Delta x}$ — the difference quotient, and everything else follows from it.

3. **Worked on $1/x$, and the cancellation is the point**
   Over a common denominator the x_0 terms cancel and one Δx divides out. Only THEN can you set $\Delta x = 0$ — before the cancellation you have $0/0$. The answer is $-1/x_0^2$: negative, and flattening as x_0 grows, both of which the graph confirms.

4. **The binomial theorem gives the power rule**
   $(x+\Delta x)^n = x^n + nx^{n-1}\Delta x + O(\Delta x^2)$. The x^n cancels, one Δx divides out, and the $O(\Delta x^2)$ junk still carries a Δx and vanishes. So $\frac{d}{dx}x^n = nx^{n-1}$, and polynomials follow term by term.

5. **Calculus gets a bad name because of the context around it**
   The calculus step in a word problem is usually one line. Everything else — intercepts, substitution, algebra — is what you already knew, and it is where the difficulty actually lives.

6. **And the deliberate sloppiness with letters**
   In the tangent-line problem y means the curve's height in one line and the equation $y = 0$ of a horizontal line two lines later. Mathematicians reuse letters rather than invent eight; you have to track which role each one is playing.

WORKED EXAMPLE

Tangent to $y = 1/x$ at (x_0, y_0), with the axes, encloses a triangle. Setting $y = 0$ gives the x-intercept $2x_0$; by the symmetry $x \leftrightarrow y$ of $xy = 1$, the y-intercept is $2y_0$. Area $= \frac{1}{2}(2x_0)(2y_0) = 2x_0y_0 = 2$ — the same for every point.

WATCH FOR

Setting $\Delta x = 0$ before simplifying. The quotient is $0/0$ until the cancellation is done; the algebra is what makes the limit exist, not a formality before it.

CHECK YOURSELF · ANSWERS AT THE BACK

1.1 The tangent line is defined as a limit of secants rather than as "the line touching once" because:

A Secants are easier to draw
B A curve can meet a line more than once and still be tangent there, so touching count does not characterise it
C Tangent lines never cross the curve
D Limits are required by convention

1.2 The triangle cut off by the axes and a tangent to $y = 1/x$ has area:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 2 OF 15

Makemore 4 — becoming a backprop ninja (DO NOT SKIP)

Sun 1 Nov – Tue 3 Nov · day 1–3 of 3 · 115 min video · Watch on YouTube

THE TAKEAWAY

Backprop is not magic to be trusted but arithmetic to be done: chunk the forward pass into small steps, push the gradient back through each by hand at the level of whole tensors, check every one against PyTorch — then collapse cross-entropy and batch norm into a single analytic backward each.

THE ARGUMENT · 8 STEPS

1 **Why do it by hand**
The 2-layer MLP with batch norm from the last lecture, its forward pass split into small named steps — embedding, linear, batch-norm mean, difference, variance with Bessel's correction (n − 1), inverse root, gain and bias, tanh, linear, then cross-entropy unrolled into max-subtraction, exp, sum, reciprocal, log. Each step can be backpropagated alone.

2 **Initialise so bugs can't hide**
b1, b2, bngain and bnbias get small random values rather than zeros or ones, because a zero can make a wrong backward formula produce the right number by accident. A cmp() helper reports, for every tensor, whether the manual gradient matches PyTorch's exactly or approximately.

3 **Exercise 1: every variable, one by one**
The loss is minus the mean of the log-probabilities at the labels, so dlogprobs is −1/n at each label and 0 elsewhere. Then: a value broadcast in the forward pass has its gradient SUMMED over the broadcast dimension in the backward; a value used twice gets the sum of both paths; a matrix multiply's gradients are matrix multiplies too, fixed by the shapes.

4 **Shapes decide matrix gradients**
For logits = h @ W2 + b2: dh = dlogits @ W2.T, dW2 = h.T @ dlogits, db2 = dlogits.sum(0). Only one arrangement of the operands produces the right shape, which is how to derive it without memorising.

5 **The embedding gradient**
emb = C[Xb] copies rows of C; going backward, each row of demb is added into dC at the index of the character it came from — and characters that appear several times in the batch accumulate.

6 **Exercise 2: cross-entropy in one line**
dlogits = softmax(logits), minus 1 at the correct class, divided by n. Each row sums to zero: the correct logit is pulled down by 1 − p and every other logit pushed up by its probability. It matches PyTorch to about 5e-9.

7 **Exercise 3: batch norm in one line**
Working through the batch-norm expression analytically gives dhprebn = bngain · bnvar_inv / n · (n · dhpreact − Σ dhpreact − n/(n−1) · bnraw · Σ (dhpreact · bnraw)) — one formula instead of nine steps, matching to about 1e-9.

8 **Exercise 4: train with your own backward**
The full training loop runs under torch.no_grad() with the hand-written gradients feeding the update — no loss.backward() at all — and reaches train 2.0705 / val 2.1099, the same quality as before.

WORKED EXAMPLE

Derive one step at a time and check it immediately with cmp(): write the local derivative, multiply by the incoming gradient, then fix the shape. If the forward pass broadcast something, sum over that dimension; if a value was used twice, add the contributions; for a matrix multiply, place the transposes so the shapes come out right.

WATCH FOR

Forgetting that a broadcast in the forward pass is a sum in the backward. counts_sum_inv is (32, 1) but multiplies a (32, 27) tensor, so its gradient must be summed over the 27 columns. Leave out the .sum(1, keepdim=True) and the gradient has the wrong shape — and later broadcasts can carry it along without an error.

CHECK YOURSELF · ANSWERS AT THE BACK

2.1 Why does the notebook initialise b1, b2, bngain and bnbias with small random values instead of the usual zeros and ones?

A Because a zero can make a wrong backward formula give the right answer
B Because zeros would make the forward pass divide by zero in batch norm
C Because random values make the network train faster to a lower loss
D Because PyTorch refuses to compute gradients for tensors that are zero

2.2 loss = −logprobs[range(n), Yb].mean() with a batch of n = 32. What is dloss/dlogprobs at a label position?

Your answer: _____

2.3 counts_sum_inv has shape (32, 1) and is multiplied by counts, of shape (32, 27). How is its gradient formed?

A By taking the gradient at the label column only, ignoring the rest
B By averaging over the 32 rows, since the batch shares one denominator
C By copying the gradient of the first column into the (32, 1) shape
D By summing (counts × dprobs) over the 27 columns it was broadcast to

SPCH 100 · STORYTELLING · LECTURE 3 OF 15

The shapes of stories — fortune drawn over time

Sun 1 Nov · 5 min video · Watch on YouTube

THE TAKEAWAY

Vonnegut draws stories as curves of fortune over time and shows that a handful of shapes — man in a hole, boy gets girl, Cinderella — are what audiences actually respond to.

THE ARGUMENT · 4 STEPS

1. **The axes**
   Up the side runs fortune, from sickness and poverty to wealth and boisterous health; along the bottom runs time, from beginning to end. It is the shape of the curve that matters.

2. **Man in a hole**
   Start a little above average, fall into trouble, climb out again. People never get sick of it.

3. **Boy gets girl**
   An ordinary day, an ordinary person; they find something wonderful, lose it, and get it back.

4. **Cinderella**
   She starts at the bottom, rises with each gift from the fairy godmother, drops at midnight — but not to the bottom, because she will remember the dance — and ends off the scale.

WORKED EXAMPLE

Before telling a story, sketch its curve on a napkin. If the line never dips, there is no trouble to get out of; if it never rises, you have only bad news. The dip-and-recovery is what holds a listener.

WATCH FOR

Telling a flat line — a pleasant sequence of events with no fall and no recovery — and wondering why nobody leans in.

CHECK YOURSELF · ANSWERS AT THE BACK

3.1 What do Vonnegut's two axes represent?

A Tension (low to high) and time (start to finish)
B Fortune (ill to good) and time (beginning to end)
C Character (weak to strong) and the length of the plot
D Fortune (poor to rich) and the number of events

3.2 Why does Cinderella not fall back to the bottom at midnight?

A The fairy godmother promises to come back for her
B She still has one glass slipper as proof of the night
C The prince starts searching for her straight away
D She will remember the dance for the rest of her life

3.3 Which shape is 'somebody gets into trouble and gets out of it again'?

A Man in a hole
B Boy gets girl
C Cinderella
D A flat line

MATH 130 · PROBABILITY · LECTURE 4 OF 15

Conditioning Continued, Law of Total Probability

Mon 2 Nov · 50 min video · Watch on YouTube

THE TAKEAWAY

Break a hard probability into disjoint cases and add them up — the law of total probability — and pair it with Bayes' rule, because the quantity you want is almost never the one you were given.

THE ARGUMENT · 13 STEPS

1 **The general strategy is decomposition**
Partition S into disjoint $A_1, \dots, A_n$ covering everything. Then $P(B) = \sum_i P(B \mid A_i)P(A_i)$. The proof is immediate from axiom 2; the ART is choosing a partition that makes each piece easy, and a bad choice turns one hard problem into n hard problems.

2 **Two cards, two conditions, two very different answers**

$P(\text{both aces} \mid \text{have an ace}) = \frac{\binom{4}{2}/\binom{52}{2}}{1-\binom{48}{2}/\binom{52}{2}} = \frac{1}{33}$. But $P(\text{both aces} \mid \text{have the ace of spades}) = \frac{3}{51} = \frac{1}{17}$ — by symmetry, the other card is equally likely to be any of 51.

3 **Why naming the suit almost doubles it**

The answer would be $1/17$ for hearts, clubs or diamonds too, so the suit is irrelevant — and yet not naming it gives $1/33$. 'At least one ace' is a union; 'the ace of spades' identifies a specific card and lets you draw the hand. The two conditions are not the same information.

4 **A test that is 95% accurate**

Interpret it: $P(T \mid D) = 0.95$ and $P(T^c \mid D^c) = 0.95$. Say which conditional you mean before computing anything — 'accurate' on its own is not a number.

5 **The patient wants the other conditional**

$P(T \mid D)$ is what the manufacturer advertises; $P(D \mid T)$ is what the patient cares about. Bayes' rule connects them: $P(D \mid T) = \frac{P(T|D)P(D)}{P(T)}$, with the denominator supplied by the law of total probability.

6 **And the answer is about 16%**

With a 1% base rate: in 1000 patients, 10 have the disease and nearly all test positive, while about 50 of the 990 healthy ones test positive too. Ten out of sixty is roughly a sixth. Asked this, most of a group of Harvard doctors guessed around 95%.

7 **Because intuition ignores the base rate**

People attend to how rarely the test errs and forget how rarely the disease occurs. The two rarities compete, and here the rarity of the disease wins.

8 **Biohazard 1: the prosecutor's fallacy**

Confusing $P(A \mid B)$ with $P(B \mid A)$. Sally Clark was convicted on $P(\text{two cot deaths} \mid \text{innocent}) = 1/73{,}000{,}000$ — a figure that assumed independence with no justification — presented as though it were $P(\text{innocent} \mid \text{evidence})$. Bayes' rule would have carried a prior for innocence that is extremely close to 1. She served years before the conviction was overturned.

9 **Biohazard 2: prior versus posterior**

Being told that A occurred does NOT mean writing $P(A) = 1$. It means $P(A \mid A) = 1$. Deciding what goes left and right of the bar is most of the work.

10 **Biohazard 3: conditional independence is a different thing**

A and B are conditionally independent given C when $P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$. Neither direction implies the other.

11 **Conditionally independent but not independent**

Games against a chess opponent of unknown strength are independent GIVEN their strength — but not unconditionally, because winning the first five tells you a great deal about the sixth.

12 **Independent but not conditionally independent**

Fire and popcorn are independent causes of the alarm. Given that the alarm sounded, ruling out popcorn makes fire certain — so conditioning on a common EFFECT creates dependence between its causes.

13 **Bayes' rule is coherent**

Update on two clues at once, or on one and then the other, or in the reverse order: the posterior is the same. Evidence can arrive in any sequence without disturbing the conclusion.

WORKED EXAMPLE

When Bayes' rule leaves you needing $P(B)$ in the denominator, that is the cue for the law of total probability. Partition on the thing you are trying to infer — disease or not, guilty or not — and the denominator becomes two easy terms with the numerator already among them.

WATCH FOR

Quoting a conditional probability without saying which way round it runs. 'The test is 95% accurate' and 'there is a 1 in 73 million chance she is innocent' are both statements whose direction was never checked, and in one case it cost someone years of their life.

CHECK YOURSELF · ANSWERS AT THE BACK

4.1 **Fire and popcorn are independent causes of an alarm. Given the alarm sounds, are they still independent?**

A Yes — independence is a property of the events themselves

B No — conditioning on a common effect makes the causes compete to explain it, so ruling one out raises the other

C Only if they are equally likely

D Yes, provided the alarm is reliable

4.2 **With a 1% base rate and a test that is 95% accurate in both directions, $P(\text{disease} \mid \text{positive})$, to three decimal places, is:**

Your answer: _____

4.3 **$P(\text{both cards are aces} \mid \text{the hand contains the ace of spades})$, to three decimal places, is:**

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 5 OF 15

Building a story live: know the end, start close to it

Mon 2 Nov – Tue 3 Nov · day 1–2 of 2 · 41 min video · Watch on YouTube

THE TAKEAWAY

Watching Dicks build a story live from a homework-for-life entry shows the craft as a chain of decisions: know the change, open on its opposite, start as late as possible, aim every beat at the end.

THE ARGUMENT · 7 STEPS

1. **Find it in the last 72 hours**
   He scans three days of entries and picks a film night with his sixteen-year-old daughter. Within 72 hours there is nearly always something worth telling.

2. **Ending first, then its opposite**
   Before telling it he names what he is trying to say — how often will moments like this happen again? — and asks what its opposite is. That becomes the beginning.

3. **Start close to the end**
   He opens during the car chase with minutes of the film left, not when they arrive at the cinema.

4. **The ending found in the telling**
   Mid-story he remembers watching her walk into the house from the car. Profound moments often happen in benign places — alone at the wheel, in a drive-through queue.

5. **End on the wish, not the hug**
   He closes hoping she will agree to a make-up cuddle. An unresolved ending keeps the story alive in the listener.

6. **Grade it kindly, then honestly**
   First what worked, then what did not: a bloated opening, too little suspense, stakes he could raise. A B minus — still better than most stories told at dinner.

7. **Anecdote or story?**
   His son's tale of smuggled dead crabs was an anecdote until a question found the change: I don't believe people; I have to learn for myself.

WORKED EXAMPLE

Ask 'what changed in me?' and write it as 'from X to Y'. The first sentence lives in X as close to the change as possible; the last lives in Y. If you cannot find an X, probe the meaning further or accept it is an anecdote.

WATCH FOR

Starting at the chronological beginning — 'so we got to the cinema'. Every sentence before the stakes appear is a sentence the audience spends waiting for a reason to care.

CHECK YOURSELF · ANSWERS AT THE BACK

5.1 What does 'start with the end' mean in Dicks's method?

A Tell the ending first, then flash back to how you got there
B Know the change you are trying to convey before you begin
C State the moral of the story in your very first sentence
D Memorise your last line so you never lose your way to it

5.2 If you cannot find an opposite of your final moment to begin with, what is most likely true?

A You have an anecdote, or have not yet found what the moment means
B The story should be told backwards, starting from the last moment
C The story needs more detail before its shape will become clear
D The final moment is too small, and a bigger one should replace it

5.3 Why does he prefer ending on the wish for a hug rather than the hug?

A The hug itself would sound too sentimental to tell
B He wasn't sure the hug ever actually happened
C Ending on the hug would make the story run too long
D An open ending keeps the story alive in the listener

MATH 110 · LINEAR ALGEBRA · LECTURE 6 OF 15

Multiplication and Inverse Matrices

Tue 3 Nov · 47 min video · Watch on YouTube

THE TAKEAWAY

Matrix multiplication is one operation with five readings, and the one that pays later is column times row, because it builds a matrix out of a single rank-one piece.

THE ARGUMENT · 6 STEPS

1 **The entry-by-entry rule is the row-column dot product**
The entry of AB in row 3, column 4 is row 3 of A dotted with column 4 of B: $\sum_k a_{3k} b_{k4}$. It is correct, it is what you compute by hand, and it says almost nothing about why the answer looks the way it does.

2 **Columns of AB are combinations of columns of A**
Take A times one column of B at a time. Each column of B is a recipe, and the column of AB it produces is that combination of A's columns. Rows of AB are, symmetrically, combinations of the rows of B.

3 **Column times row makes a whole matrix**

A column of A times a row of B — an $m \times 1$ times a $1 \times n$ — is a full $m \times n$ matrix. Every one of its columns is a multiple of that one column, and every row a multiple of that one row: a rank-one matrix.

4 **AB is the sum of those rank-one pieces**

Column k of A times row k of B, summed over k. The whole product is a stack of rank-one layers. That decomposition comes back for the four subspaces and again, in its best form, at the SVD.

5 **Blocks multiply by the same rule**

Cut both matrices into blocks that fit, and the block in position 1,1 of the product is $A_1B_1 + A_2B_3$ — the row-column rule again, with matrices where the numbers were.

6 **Gauss-Jordan solves for every column of A^{-1} at once**

$AA^{-1} = I$ is n systems sharing one matrix. Augment $[A; |; I]$ and eliminate downwards and then upwards until the left half is I. Whatever E did that satisfies $EA = I$, so E IS A^{-1} — and it has been applied to I on the right, leaving A^{-1} sitting there.

WORKED EXAMPLE

Invert $\begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$. Augment with I, subtract twice row one from row two to get a 1 in the corner, then subtract three times the new row two from row one. The right half reads $\begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix}$.

WATCH FOR

Calling a matrix singular because its determinant happened to be zero. The reason is that some non-zero $\vec{x}$ has $A\vec{x} = \vec{0}$; an inverse would multiply that to give $\vec{x} = \vec{0}$, which is false.

KEY CONCEPTS

Multiplication is composition

CHECK YOURSELF · ANSWERS AT THE BACK

6.1 A column of A times a row of B produces:

A A single number
B A full matrix, every column a multiple of that one column
C A vector
D The identity

6.2 The inverse of $\begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$ has, in row 1 column 2, the entry:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 7 OF 15

Limits

Wed 4 Nov · 53 min video · Watch on YouTube

THE TAKEAWAY

A derivative is a rate of change, and making that rigorous means separating left and right limits — which is also what makes continuity definable.

THE ARGUMENT · 6 STEPS

1. **The same ratio, read as a rate**
   $\Delta y/\Delta x$ is the AVERAGE rate of change over the interval; dy/dx is the instantaneous one. Charge gives current, distance gives speed, temperature gives the gradient that drives weather.

2. **The pumpkin drop makes the difference visceral**
   From 80 m, $h = 80 - 5t^2$ reaches the ground at $t = 4$. Average speed is -20 m/s. But $h' = -10t$, so at impact it is -40 m/s — twice the average, about 90 mph. Nobody at the event cares about the average.

3. **Sensitivity of measurement is the third use**
   GPS measures a distance h with some error Δh and deduces a horizontal distance L. How bad the error in L is comes from $\Delta L/\Delta h \approx dL/dh$. That is why derivatives matter for landing aircraft.

4. **Easy limits versus the limits derivatives need**
   $\lim_{x\to 4} \frac{x+3}{x^2+1}$ is done by substituting. A derivative NEVER is: plugging in gives $0/0$ every single time, so some cancellation is always required first.

5. **Left and right limits, and the zoo of discontinuities**
   A jump has both one-sided limits existing but unequal. A removable one has them equal with a hole — $\frac{\sin x}{x}$ at 0 is the important case. An infinite one splits: $1/x$ goes to $+\infty$ from the right and $-\infty$ from the left, so writing a single limit there is simply wrong.

6. **Differentiable implies continuous, in one line**
   $\lim(f(x) - f(x_0)) = \lim \frac{f(x)-f(x_0)}{x-x_0} \cdot (x - x_0) = f'(x_0) \cdot 0 = 0$. Multiplying and dividing by $x - x_0$ is legal precisely because the limit never evaluates AT x_0, so that factor is small but never zero.

WORKED EXAMPLE

The derivative graph looks nothing like the function's. $1/x$ has two branches, one rising and one falling; $-1/x^2$ is negative everywhere and dives to $-\infty$ on both sides. What it plots is the SLOPE, and slope has its own shape.

WATCH FOR

Writing $\lim_{x\to 0} \frac{1}{x} = \infty$. The two sides disagree: $+\infty$ from the right and $-\infty$ from the left. Specify the side, or say the limit does not exist.

CHECK YOURSELF · ANSWERS AT THE BACK

7.1 Multiplying and dividing by $(x - x_0)$ inside a limit is legal because:

A Zero times anything is zero
B The limit never evaluates at $x = x_0$, so that factor is always non-zero
C Limits ignore algebra
D x_0 is always positive

7.2 For $h = 80 - 5t^2$, the speed at impact ($t = 4$) in metres per second is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 8 OF 15

Makemore 5 — building a WaveNet

Wed 4 Nov – Thu 5 Nov · day 1–2 of 2 · 56 min video · Watch on YouTube

THE TAKEAWAY

Seeing more context helps, but squashing eight characters into one layer is crude: fuse them a pair at a time — a tree, as in WaveNet — with PyTorch-style modules, and mind the shapes, because one wrong reduction dimension in batch norm silently costs performance.

THE ARGUMENT · 7 STEPS

1. **A longer context**
   block_size goes from 3 to 8, so every example is the previous eight characters ('..yuheng' predicts '.'). On its own this improves the dev loss from 2.105 to 2.027.

2. **Modules like torch.nn**
   Linear (Kaiming-scaled), BatchNorm1d, Tanh, plus new Embedding, FlattenConsecutive and Sequential classes — a model becomes a list of layers, and model(Xb) gives the logits. Layers are switched to eval mode before evaluating, which batch norm needs.

3 **Flatten a pair at a time**

FlattenConsecutive(n) reshapes (B, T, C) into (B, T/n, C·n) — joining n neighbouring positions — and squeezes away a dimension of size 1. Three FlattenConsecutive(2) blocks fuse 8 characters into 4 pairs, then 2 groups of four, then one: a tree, like WaveNet, instead of one layer seeing all eight at once.

4 **Batch norm on 3-D input**

Inside the tree the activations are (B, T, C). The batch statistics must be taken over dims (0, 1) — every example and every position — so each of the C channels gets one mean and variance. Over dim 0 alone, each position gets its own, from fewer numbers; fixing that is the 'fix bug in batchnorm' line in the log.

5 **What helped, by the log**

Context 3 to 8: val 2.105 to 2.027. Flat to hierarchical at the same 22K parameters: 2.029, no gain by itself. The batch-norm fix: 2.022. Scaling to n_embd 24 and n_hidden 128 (76,579 parameters): train 1.769, val 1.993.

6 **Samples, and the gap**

The names now read well ('hendrix.', 'jaylene.', 'aubreana.'). But train 1.769 against val 1.993 is a wider gap than before: the bigger model is starting to overfit.

7 **Why convolutions, next**

Forwarding the eight examples of 'diondre' one at a time repeats work: a convolution is 'a for loop' that applies the same Linear layers across every position in one efficient pass, reusing the shared intermediate results.

WORKED EXAMPLE

Push one batch through the model layer by layer and print each layer's output shape: (32, 8) indices, (32, 8, 24) embeddings, (32, 4, 48) after the first flatten, (32, 4, 128) after its linear layer, and so on down to (32, 27). Any reduction — a mean, a variance — should be checked against those shapes, which is how the batch-norm bug shows itself.

WATCH FOR

Copying a 2-D layer into a 3-D setting unchanged. BatchNorm1d written for (B, C) still runs on (B, T, C), but averaging over dim 0 alone gives every position its own statistics — no error, just a quietly worse model.

CHECK YOURSELF · ANSWERS AT THE BACK

8.1 The hierarchical model: Embedding(27, 24); then three blocks of FlattenConsecutive(2) and Linear without bias (48→128, 256→128, 256→128), each with BatchNorm1d(128); then Linear(128, 27) with bias. How many parameters? (digits only, no commas)

Your answer: _____

8.2 Why does the hierarchical model apply FlattenConsecutive(2) three times instead of flattening all 8 characters into the first layer?

A Because a Linear layer cannot take more than 48 inputs at a time
B Because flattening all 8 at once would make batch norm impossible
C Because the embedding only works on pairs of characters at a time
D To fuse the context gradually, pairs then pairs of pairs, as a tree

8.3 Inside the tree, batch norm sees inputs of shape (32, 4, 128). Why must its mean and variance be over dims (0, 1) rather than 0?

A Over dim 0 the result has the wrong number of dimensions and errors
B Over (0, 1) is faster to compute, though both give the same answer
C So each of the 128 channels gets one estimate, from all 4 positions
D So the running mean can be skipped while the model is training

SPCH 100 · STORYTELLING · LECTURE 9 OF 15

And, But, Yet, Therefore — structure in one breath

Wed 4 Nov · 4 min video · Watch on YouTube

THE TAKEAWAY

Tamsen Webster takes Randy Olson's And, But, Therefore — the three acts of a story in three words — and adds a Yet for the moment of truth that makes the resolution persuasive.

THE ARGUMENT · 4 STEPS

1. **And**
   Two pieces of information that set the scene: stories are powerful, and there is plenty of advice about them.

2. **But**
   The problem: we rarely have time to apply that advice to our messages.

3. **Therefore**
   The resolution: we need simpler ways to give messages a story's shape.

4 **Yet**

The moment of truth inside the second act — the extra fact that makes the audience agree with the Therefore. Here: the easier something is, the more likely we are to use it.

WORKED EXAMPLE

When you have ten seconds to prepare, say four lines: the situation (And), the problem (But), the fact that changes their mind (Yet), and what should happen (Therefore). It is a story's skeleton without the story.

WATCH FOR

A Yet that is just another restatement of the problem. If deleting it changes nothing, the audience has not been given a reason to accept your Therefore.

CHECK YOURSELF · ANSWERS AT THE BACK

9.1 Which act of a story does the 'But' stand for?

A The setup, or background
B The conflict, or problem
C The resolution, or fix
D The moral, or takeaway

9.2 What does Webster's added 'Yet' supply?

A A second problem that raises the stakes just before the resolution
B A joke that relaxes the audience before the hard part arrives
C The moment of truth that makes the audience accept the resolution
D A summary of the setup, so the audience remembers the context

9.3 When is this structure most useful?

A When the audience already agrees with you and needs no persuading
B When you must shape a message quickly, with no time for a full story
C When the story runs longer than ten minutes and needs firm signposts
D When you are writing a report rather than speaking to an audience

SPCH 100 · STORYTELLING · LECTURE 10 OF 15

Structure is what they know, and when — the Nemo flashbacks

Wed 4 Nov · 4 min video · Watch on YouTube

THE TAKEAWAY

Pixar story artist Kristen Lester: structure is what you want the audience to know, and when. A mushroom joke told out of order dies, and Finding Nemo's audiences only warmed to Marlin once his loss moved from late flashbacks to the opening.

THE ARGUMENT · 4 STEPS

1. **The mushroom joke**
   Opening, build-up, punchline ('I'm a fungi'). Botch the opening or the middle, or let every lady say yes, and there's no joke.

2. **What, and when**
   Structure decides what the audience knows at each moment, and so what they feel.

3. **Nemo's flashbacks**
   Early versions wove Marlin and Coral's past through the film and revealed the barracuda attack near the end. Audiences didn't like Marlin.

4. **The fix**
   Almost all the flashbacks went; the attack moved to the beginning. Knowing his loss from the start, audiences saw his protectiveness as love and his journey as brave.

WORKED EXAMPLE

Same events, new order: once the audience sees Coral and all but one of the eggs taken in the opening minutes, 'I promise I will never let anything happen to you' explains every over-cautious thing Marlin does afterwards.

WATCH FOR

This is the short opening video of a longer Khan Academy lesson; the exercises are on Khan Academy's Pixar in a Box pages.

CHECK YOURSELF · ANSWERS AT THE BACK

10.1 How does Lester define structure?

A The number of acts and where each one ends
B What you want the audience to know, and when
C The order in which the events really happened
D How long each scene runs before the next

10.2 Why didn't early audiences like Marlin?

A They didn't learn why he was so protective until near the end
B His fussing over Nemo made the film feel slow and repetitive
C They saw the barracuda attack too early and found it upsetting
D Dory was funnier, so audiences took her side against him

10.3 What breaks the mushroom joke in her last failed attempt?

A A punchline given away too early
B An opening that skips the setup
C No conflict — everyone says yes
D A build-up told much too fast

SPCH 100 · STORYTELLING · LECTURE 11 OF 15

Write what you know — the feeling under the monster story

Wed 4 Nov · 3 min video · Watch on YouTube

THE TAKEAWAY

Pete Docter opens Pixar in a Box's storytelling season: stories connect on an emotional level, 'write what you know' means putting your own feelings into any plot, and stories only sparkle after many retellings.

THE ARGUMENT · 4 STEPS

1. **'What happened?'**
   When Val asks Pete what happened, she is asking for a story. People have told them for as long as they could speak.

2. **Write what you know**
   As a kid he wanted explosions and car chases, not suburban Minnesota. You can have both — put how you felt into the explosions.

3. **Monsters, Inc.**
   'A monster who scares kids for a living' got smiles in a pitch but left audiences restless. It was really about a man becoming a father — what was happening to Docter.

4. **Retell it**
   Not even the greats get it right first time; Pixar's stories improve over retelling after retelling, up to the thirtieth and beyond.

WORKED EXAMPLE

Monsters, Inc.: the clock-in, eat-donuts, talk-about-union-dues joke was funny for a minute. The film found its spine when Docter recognised his own life in it — someone suddenly responsible for a small child.

WATCH FOR

Under three minutes of introduction; the rest of the season is on Khan Academy. Its value here is one move: name the feeling under your plot.

CHECK YOURSELF · ANSWERS AT THE BACK

11.1 What does 'write what you know' mean in Docter's telling?

A Write only about places and people you know well
B Put something you have felt into whatever you write
C Avoid fantasy and stick to events that really happened
D Research every detail until you know it as an expert

11.2 What did Docter realise Monsters, Inc. was really about?

A A man becoming a father
B Union politics
C Fear of the dark
D A friendship between monsters

11.3 What does he say about getting a story right?

A The best storytellers get it right first time
B Two drafts are enough if the idea is strong
C It takes many retellings before it sparkles
D Test it once on an audience, then lock it

MATH 130 · PROBABILITY · LECTURE 12 OF 15

Monty Hall, Simpson's Paradox

Thu 5 Nov · 49 min video · Watch on YouTube

THE TAKEAWAY

Condition on ALL the evidence — in Monty Hall the evidence is not that door 2 has a goat but that Monty opened it — and remember that an inequality holding in every stratum can reverse when the strata are pooled.

THE ARGUMENT · 14 STEPS

1. **State the assumptions, because the answer depends on them**
One car, two goats, all equally likely; Monty KNOWS where the car is; Monty always opens a goat door and always offers the switch; and when he has a choice he picks either with probability $\frac{1}{2}$. Change any one and the problem changes.

2. **The naive answer, and its flaw**
Two doors left, no reason to prefer either, so 50-50. That is the naive definition applied without checking that the two are equally likely — and conditionally they are not.

3. **What you must condition on**
Not 'door 2 has a goat' but 'Monty opened door 2'. Door 2 having a goat was always likely; Monty CHOOSING it is the informative part, because his hand was sometimes forced.

4 **The tree settles it**

Car behind 1 (prob $\frac{1}{3}$): Monty picks 2 or 3, each $\frac{1}{2}$. Car behind 2: he must open 3. Car behind 3: he must open 2. Condition on his opening 2, keep the two surviving paths — weights $\frac{1}{6}$ and $\frac{1}{3}$ — renormalise to $\frac{1}{3}$ and $\frac{2}{3}$.

5 **Or wish you knew where the car was**

The law of total probability is wishful thinking made rigorous: condition on the thing you wish you knew. $P(S) = 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3}$, because switching wins exactly when the first guess was wrong.

6 **The conditional and unconditional answers agree here**

By symmetry between doors 2 and 3, $P(S \mid \text{Monty opens } 2) = P(S) = \frac{2}{3}$. Break the symmetry — a lazy Monty who prefers door 2 — and they come apart, though the unconditional answer stays $\frac{2}{3}$.

7 **A million doors**

Pick one of a million, watch Monty open 999,998 goat doors, and nobody declines to switch. The argument is identical to the three-door case; only the intuition improves.

8 **And it is trivially simulable**

The controversy after Marilyn vos Savant's column drew thousands of angry letters, some from mathematics PhDs. Anyone could have settled it in an afternoon with three cups.

9 **Simpson's paradox**

Dr Hibbert beats Dr Nick at heart surgery (70/90 against 2/10) and at bandage removal (10/10 against 81/90) — and loses overall, 80% to 83%. Every conditional comparison points one way; the aggregate points the other.

10 **Because the case mix differs**

Ninety per cent of Dr Nick's work is the easy surgery. The world's leading neurosurgeon gets the cases nobody else will take, and their raw success rate suffers for it.

11 **In notation**

$P(A \mid B, C) < P(A \mid B^c, C)$ and $P(A \mid B, C^c) < P(A \mid B^c, C^c)$, yet $P(A \mid B) > P(A \mid B^c)$. C is the CONFOUNDER — the thing you should have controlled for.

12 **Why the inequalities do not combine**

The conditional law of total probability gives $P(A \mid B) = P(A \mid B, C)P(C \mid B) + P(A \mid B, C^c)P(C^c \mid B)$. The weights are conditional on B, and the corresponding weights for B^c are different numbers. Nothing relates them, so the inequality does not carry through.

13 **It is the same as adding fractions wrongly**

Aggregating counts is $\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d}$, which is not how fractions add. If it were, Simpson's paradox could not happen.

14 **And it reaches the courts**

Berkeley's graduate admissions looked discriminatory in aggregate while almost no individual department did, because women applied disproportionately to the departments that admit fewest.

WORKED EXAMPLE

When a problem stalls, ask what you wish you knew and condition on it. In Monty Hall that is the car's location; in the disease test it is whether the patient is ill. The law of total probability turns the wish into a legitimate step.

WATCH FOR

Conditioning on the fact rather than on how you learned it. 'Door 2 has a goat' and 'Monty opened door 2' are different pieces of evidence, and only the second carries information about door 3.

CHECK YOURSELF · ANSWERS AT THE BACK

12.1 Why is the answer to Monty Hall not 1/2?

A Because the doors were never equally likely
B Because the evidence is that Monty CHOSE to open door 2, and his choice was constrained by where the car is
C Because Monty might be lying
D Because there are three doors, not two

12.2 The probability of winning Monty Hall by switching, to three decimal places, is:

Your answer: _____

12.3 Dr Nick's overall success rate across both surgeries, as a proportion, is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 13 OF 15

Credibility before vulnerability — five tips for a story in a talk

Thu 5 Nov · 7 min video · Watch on YouTube

THE TAKEAWAY

Broadcaster Riaz Meghji's five tips for telling a story to an audience: know its purpose, open on an unexpected truth, speak to the senses, move from struggle to conflict to resolution with a takeaway, and show credibility before vulnerability.

THE ARGUMENT · 5 STEPS

1. **Purpose**
   Inform, persuade, entertain? Frame a problem the audience cares about now. He borrows John Maxwell's four ingredients: hope, help, heart and humour.

2. **An unexpected truth**
   'Last month I received an email I was never supposed to see.' A secret with a silver lining sets up a change.

3. **The senses**
   What did you see, hear, smell, touch, taste? Detail draws the audience into the scene with you.

4. **Struggle, conflict, resolution**
   People cheer for an underdog only after they relate to the struggle; the conflict builds suspense; the resolution says how you changed and what they can take from it.

5. **Credibility before vulnerability**
   The pratfall effect: a reveal draws people closer only if they already see you as competent.

WORKED EXAMPLE

In a pitch: first one line of credibility ('I've rebuilt the books for forty firms like yours'), then the vulnerable story ('the first one, I nearly lost because I missed something obvious'), then what it taught you — and what that means for them.

WATCH FOR

Short and general; several tips echo earlier lessons in this unit (the elephant, the five-second moment). Its distinctive contribution is the order: credibility first, then vulnerability.

CHECK YOURSELF · ANSWERS AT THE BACK

13.1 What does 'credibility before vulnerability' mean?

A Share a flaw first, because vulnerability is what builds your credibility
B Show you're competent before you share something raw, or it can backfire
C Only tell personal stories once you're an expert in the field
D Be credible by keeping vulnerable stories out of professional talks

13.2 Which kind of opening does he recommend?

A An unexpected truth or a secret
B A striking statistic or a fact
C A question put to the audience
D A dictionary definition of the topic

13.3 What three stages does he suggest in place of beginning, middle and end?

A Hook, story, call to action
B Problem, solution, and the benefit
C Struggle, conflict, resolution
D Past, present, and the future

MATH 110 · LINEAR ALGEBRA · LECTURE 14 OF 15

Factorization into A = LU

Fri 6 Nov · 48 min video · Watch on YouTube

THE TAKEAWAY

Elimination undone is tidier than elimination done: the multipliers drop straight into L, so the factorisation $A = LU$ costs nothing beyond the elimination already performed.

THE ARGUMENT · 6 STEPS

1. **Inverses and transposes reverse the order**
   $(AB)^{-1} = B^{-1}A^{-1}$ — shoes off, then socks, so socks back on before shoes. Transposing $AA^{-1} = I$ gives the same reversal and a second fact: $(A^T)^{-1} = (A^{-1})^T$, so the two operations commute on a single matrix.

2. **Move the elimination matrices to the other side**
   $E_{32}E_{31}E_{21}A = U$. Multiply back by the inverses, in the opposite order, and $A = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U$. Name that lower triangular product L and you have $A = LU$.

3 **The forward product invents a number nobody chose**

Multiply E_{21} (subtract 2 of row one from row two) by E_{32} (subtract 5 of row two from row three) and a 10 appears in the 3,1 corner: ten of row one reached row three by going through row two.

4 **The reverse product does not**

Take the same two steps as inverses in the opposite order and the entries stay where they were put: 2 and 5 on the diagonal's south side, nothing between them. L is a record, not a computation.

5 **So A can be thrown away**

Once a row of U and the multipliers that made it are saved, the original row has nothing left to say. L and U together are the matrix, held in the same amount of storage.

6 **The price is $n^3/3$ once, then n^2 per right-hand side**

Clearing the first column touches about n^2 entries, the next about $(n-1)^2$, and the sum behaves like $\int x^2\,dx$. That is why you factor once and then run each new $\vec{b}$ through cheaply.

WORKED EXAMPLE

$A = \begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}$. The multiplier is 4, and subtracting 4 times row one from row two leaves $U = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. L is the identity carrying that same 4 below the diagonal, and LU returns A.

WATCH FOR

Multiplying the elimination matrices together to find L. That product mixes multipliers — a 2 and a 5 become a 10 — while their inverses in reverse order leave them alone.

KEY CONCEPTS

Multiplication is composition

CHECK YOURSELF · ANSWERS AT THE BACK

14.1 The multipliers appear undisturbed in L, but not in the product of the Es, because:

A L is upper triangular
B In the reverse order, no later step feeds an earlier row's multiple further down
C The Es are not invertible
D L has the pivots on its diagonal

14.2 With E_{21} subtracting 2 of row one from row two and E_{32} subtracting 5 of row two from row three, the entry of $E_{32}E_{21}$ in row 3, column 1 is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 15 OF 15

Let's build GPT — from scratch, in code

Fri 6 Nov · day 1 of 3 · 116 min video · Watch on YouTube · finishes in week 4

THE TAKEAWAY

A GPT is a bigram model that learned to look back: tokens gather information from earlier positions through masked, scaled self-attention, compute on it with a feed-forward net, and stack those blocks with residual connections and layer norms — each addition visibly lowering the validation loss on Tiny Shakespeare.

THE ARGUMENT · 8 STEPS

1. **Characters as tokens, chunks as batches**
   Tiny Shakespeare is encoded one character per integer, split 90 / 10 into train and val. get_batch draws random chunks: x is block_size tokens and y the same chunk shifted by one, so one chunk holds block_size examples, with contexts of length 1 up to block_size.

2. **The baseline: a bigram model**
   Each token reads the logits for the next token straight out of an embedding table; generate() samples one token at a time from the last position's softmax and appends it. Trained with AdamW, it produces Shakespeare-shaped gibberish.

3. **Embeddings with positions**
   Tokens get n_embd-dimensional embeddings and a linear lm_head to the vocabulary; a position embedding table adds where each token sits, since attention by itself has no notion of order.

4. **One head of self-attention**
   Each position emits a query and a key; their dot products, scaled by $1/\sqrt{\text{head size}}$, are the affinities (the gpt.py published with the video scaled by the embedding size C instead; a fix after the video changed it). A lower-triangular mask sets the future to −inf, softmax turns each row into weights, and the output is a weighted sum of the values. Val loss: 2.5.

5. **Many heads, then computation**
   Four heads of size 8, concatenated back to 32: several independent communication channels (val 2.28). A per-position feed-forward net — linear, ReLU — then lets each token compute on what it gathered (2.24). Attention is communication; the feed-forward net is computation.

6. **Residuals and layer norm make it deep**
   Blocks are stacked with x = x + f(x), a projection back into the residual stream, and a 4x wider feed-forward layer; layer norm is applied to x before each sub-layer and once more before lm_head. Val loss 2.06.

7 **Scaling up**

batch 64, block_size 256, n_embd 384, 6 heads, 6 layers, learning rate 3e-4 and dropout 0.2 on the attention weights, the projections and the feed-forward output — and the samples in more.txt read like a play.

8 **Generation has a window**

generate() crops the running sequence to its last block_size tokens before every forward pass: the position table has only block_size rows, and the mask is that wide.

WORKED EXAMPLE

Track the shapes through one head: x is (B, T, C); q and k are (B, T, hs); q @ k.transpose(-2, -1) is (B, T, T) — one row of affinities per position; after the mask and softmax, @ v gives (B, T, hs). Then multi-head concatenates n_head of those back to C, which is why head_size = n_embd // n_head.

WATCH FOR

Forgetting the mask, or masking after the softmax. Without the −inf entries in the upper triangle, every position can attend to the tokens it is supposed to predict, the training loss collapses, and generation — which has no future to look at — falls apart.

CHECK YOURSELF · ANSWERS AT THE BACK

15.1 get_batch takes x = data[i : i+block_size] and y = data[i+1 : i+block_size+1]. With block_size = 8, how many (context, target) training examples does one such chunk pack?

Your answer: _____

15.2 In a self-attention head, why is wei filled with −inf wherever the lower-triangular tril matrix is 0, before the softmax?

A To stop the attention weights from growing too large during training
B To make the attention weights symmetric between any two positions
C So each position gathers only from itself and earlier positions, never later
D So that the softmax output will sum to the head size, rather than to exactly one

15.3 The lecture scales the attention scores by 1/√(head size), so that they keep roughly unit variance before the softmax. With n_embd = 384 and n_head = 6, by what factor are the scores q · k multiplied? Three decimals.

Your answer: _____

WEEK 3 · SUN 1 – FRI 6 NOVEMBER 2026

What this week established

Each lecture’s takeaway, by course.

MATH 120 · Single-Variable Calculus

1 **Rate of Change.** The derivative is the slope of the tangent line, defined as the limit of secant slopes — and the difference quotient is the formula that makes it computable.

7 **Limits.** A derivative is a rate of change, and making that rigorous means separating left and right limits — which is also what makes continuity definable.

AI 200 · Neural Networks: Zero to Hero

2 **Makemore 4 — becoming a backprop ninja (DO NOT SKIP).** Backprop is not magic to be trusted but arithmetic to be done: chunk the forward pass into small steps, push the gradient back through each by hand at the level of whole tensors, check every one against PyTorch — then collapse cross-entropy and batch norm into a single analytic backward each.

8 **Makemore 5 — building a WaveNet.** Seeing more context helps, but squashing eight characters into one layer is crude: fuse them a pair at a time — a tree, as in WaveNet — with PyTorch-style modules, and mind the shapes, because one wrong reduction dimension in batch norm silently costs performance.

15 **Let's build GPT — from scratch, in code.** A GPT is a bigram model that learned to look back: tokens gather information from earlier positions through masked, scaled self-attention, compute on it with a feed-forward net, and stack those blocks with residual connections and layer norms — each addition visibly lowering the validation loss on Tiny Shakespeare.

SPCH 100 · Storytelling

3 **The shapes of stories — fortune drawn over time.** Vonnegut draws stories as curves of fortune over time and shows that a handful of shapes — man in a hole, boy gets girl, Cinderella — are what audiences actually respond to.

5 **Building a story live: know the end, start close to it.** Watching Dicks build a story live from a homework-for-life entry shows the craft as a chain of decisions: know the change, open on its opposite, start as late as possible, aim every beat at the end.

9 **And, But, Yet, Therefore — structure in one breath.** Tamsen Webster takes Randy Olson's And, But, Therefore — the three acts of a story in three words — and adds a Yet for the moment of truth that makes the resolution persuasive.

10 **Structure is what they know, and when — the Nemo flashbacks.** Pixar story artist Kristen Lester: structure is what you want the audience to know, and when. A mushroom joke told out of order dies, and Finding Nemo's audiences only warmed to Marlin once his loss moved from late flashbacks to the opening.

11 **Write what you know — the feeling under the monster story.** Pete Docter opens Pixar in a Box's storytelling season: stories connect on an emotional level, 'write what you know' means putting your own feelings into any plot, and stories only sparkle after many retellings.

13 **Credibility before vulnerability — five tips for a story in a talk.** Broadcaster Riaz Meghji's five tips for telling a story to an audience: know its purpose, open on an unexpected truth, speak to the senses, move from struggle to conflict to resolution with a takeaway, and show credibility before vulnerability.

MATH 130 · Probability

4 **Conditioning Continued, Law of Total Probability.** Break a hard probability into disjoint cases and add them up — the law of total probability — and pair it with Bayes' rule, because the quantity you want is almost never the one you were given.

12 **Monty Hall, Simpson's Paradox.** Condition on ALL the evidence — in Monty Hall the evidence is not that door 2 has a goat but that Monty opened it — and remember that an inequality holding in every stratum can reverse when the strata are pooled.

MATH 110 · Linear Algebra

6 **Multiplication and Inverse Matrices.** Matrix multiplication is one operation with five readings, and the one that pays later is column times row, because it builds a matrix out of a single rank-one piece.

14 **Factorization into A = LU.** Elimination undone is tidier than elimination done: the multipliers drop straight into L, so the factorisation $A = LU$ costs nothing beyond the elimination already performed.

WEEK 3 · SUN 1 – FRI 6 NOVEMBER 2026

Answers

The answers and explanations are the summaries' own.

1 Rate of Change

1.1 B — A curve can meet a line more than once and still be tangent there, so touching count does not characterise it

A wiggly curve may cross its tangent line again elsewhere. What actually pins the tangent down is that secant slopes through a nearby point converge to it as that point approaches.

1.2 2

The intercepts are $2x_0$ and $2y_0$, so the area is $\frac{1}{2}(2x_0)(2y_0) = 2x_0y_0$. On this curve $x_0y_0 = 1$, so the area is 2 regardless of where the tangent touches.

2 Makemore 4 — becoming a backprop ninja (DO NOT SKIP)

2.1 A — Because a zero can make a wrong backward formula give the right answer

The notebook says so: initialising with e.g. all zeros could mask an incorrect implementation of the backward pass. If a term in your derivative is multiplied by a parameter that happens to be 0, a mistake in that term vanishes and cmp() reports a match.

2.2 −0.03125

The loss is minus the average of 32 of the entries of logprobs, so each of those entries gets $-1/32 = -0.03125$, and every other entry — which the loss never reads — gets 0. That is exactly how dlogprobs is built: zeros, then −1.0/n at the labels.

2.3 D — By summing (counts × dprobs) over the 27 columns it was broadcast to

Broadcasting reused each row's single value 27 times, so by the multivariable chain rule its gradient is the sum of the 27 contributions: dcounts_sum_inv = (counts * dprobs).sum(1, keepdim=True). A broadcast forward is a sum backward.

3 The shapes of stories — fortune drawn over time

3.1 B — Fortune (ill to good) and time (beginning to end)

The G-I axis is good fortune to ill fortune; the B-E axis is beginning to end.

3.2 D — She will remember the dance for the rest of her life

The loss is real but she ends above where she began, because the memory stays.

3.3 A — Man in a hole

Vonnegut says people never tire of it.

4 Conditioning Continued, Law of Total Probability

4.1 B — No — conditioning on a common effect makes the causes compete to explain it, so ruling one out raises the other

If the alarm sounds and nobody is making popcorn, fire becomes certain. Unconditional independence says nothing about what happens once you condition on something both causes influence.

4.2 0.161

$\frac{0.95\times0.01}{0.95\times0.01+0.05\times0.99} = \frac{0.0095}{0.059} = 0.161$. The 49.5 false positives per thousand swamp the 9.5 true ones.

4.3 0.059

$3/51 = 1/17 = 0.0588$. Nearly double the $1/33 = 0.0303$ you get from the weaker condition 'the hand contains an ace'.

5 Building a story live: know the end, start close to it

5.1 B — Know the change you are trying to convey before you begin

You cannot aim at an ending you have not chosen; the change you want to convey decides everything else.

5.2 A — You have an anecdote, or have not yet found what the moment means

His son's crab story became a story only once a question uncovered what it meant.

5.3 D — An open ending keeps the story alive in the listener

People wonder what happened next, and the story keeps working in them.

6 Multiplication and Inverse Matrices

6.1 B — A full matrix, every column a multiple of that one column

An $m \times 1$ times a $1 \times n$ is $m \times n$. Its column space is one line and its row space is one line — rank one — and AB is the sum of n such layers.

6.2 −3

The determinant is $7 - 6 = 1$, so the inverse is $\begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix}$ and the 1,2 entry is -3. Check it: $1 \cdot (-3) + 3 \cdot 1 = 0$, as the off-diagonal of I requires.

7 Limits

7.1 B — The limit never evaluates at $x = x_0$, so that factor is always non-zero

A limit as $x \to x_0$ deliberately excludes $x = x_0$ itself. The factor is arbitrarily small but never zero, so the division is valid at every point the limit actually looks at.

7.2 −40

$h'(t) = -10t$, so $h'(4) = -40$. The average speed over the fall was only -20, since the pumpkin started at rest.

8 Makemore 5 — building a WaveNet

8.1 76579

648 (embedding) + 6,144 + 256 + 32,768 + 256 + 32,768 + 256 (three linear + batch-norm pairs; each batch norm has gamma and beta) + 3,456 + 27 (output) = 76,579, as the notebook prints. The 256-wide inputs are two fused 128-wide positions.

8.2 D — To fuse the context gradually, pairs then pairs of pairs, as a tree

Flattening all eight into one layer crushes the whole context in a single step. Joining two neighbours at a time and then joining those results builds a tree — the WaveNet idea — so information is combined gradually across depth. The commented 'original network' is the flat version.

8.3 C — So each of the 128 channels gets one estimate, from all 4 positions

Averaging over dim 0 only gives a (1, 4, 128) mean: separate statistics per position, each from 32 numbers, and running buffers of that shape. It runs without error, which is why it is a bug. Over (0, 1) every channel gets one mean and variance from 128 numbers — the 'fix bug in batchnorm' entry in the log.

9 And, But, Yet, Therefore — structure in one breath

9.1 B — The conflict, or problem

And is setup, But is conflict, Therefore is resolution.

9.2 C — The moment of truth that makes the audience accept the resolution

It is the extra piece of information that persuades the audience the Therefore follows.

9.3 B — When you must shape a message quickly, with no time for a full story

She presents it as the fast way to apply story structure on the fly.

10 Structure is what they know, and when — the Nemo flashbacks

10.1 B — What you want the audience to know, and when

Get the order wrong and the effect changes, even with the same material.

10.2 A — They didn't learn why he was so protective until near the end

The flashback structure hid his reason until it was too late.

10.3 C — No conflict — everyone says yes

Without conflict, joke over.

11 Write what you know — the feeling under the monster story

11.1 B — Put something you have felt into whatever you write

Something from your own life makes the story come alive.

11.2 A — A man becoming a father

That was what was happening to him at the time.

11.3 C — It takes many retellings before it sparkles

Up to the thirtieth time, and on.

12 Monty Hall, Simpson's Paradox

12.1 B — Because the evidence is that Monty CHOSE to open door 2, and his choice was constrained by where the car is

When the car is behind door 3 he had no choice; when it is behind door 1 he had two. That asymmetry is the information, and conditioning only on 'door 2 has a goat' discards it.

12.2 0.667

$2/3$. Switching wins exactly when the first guess was wrong, which happens two times in three.

12.3 0.83

$(2+81)/100 = 0.83$, against Dr Hibbert's 0.80 — even though Hibbert wins both individual comparisons. Ninety per cent of Nick's cases are the easy operation.

13 Credibility before vulnerability — five tips for a story in a talk

13.1 B — Show you're competent before you share something raw, or it can backfire

How a reveal lands depends on how they saw you beforehand.

13.2 A — An unexpected truth or a secret

It makes the audience ask what happened next.

13.3 C — Struggle, conflict, resolution

Relate to the struggle, build suspense in the conflict, reveal the change.

14 Factorization into A = LU

14.1 B — In the reverse order, no later step feeds an earlier row's multiple further down

Forwards, a multiple of row one is put into row two and then carried on into row three. Taking the inverses in the opposite order never composes the two steps, so each multiplier stays in its own slot.

14.2 10

$(-5)\cdot(-2) = 10$: two of row one went into row two, and five copies of that new row two went into row three.

15 Let's build GPT — from scratch, in code

15.1 8

Position t of x, with everything before it, predicts y[t]: contexts of length 1, 2, ... up to 8, each with its own target. Training on every length is also what lets the model generate from a context as short as one token.

15.2 C — So each position gathers only from itself and earlier positions, never later

exp(–inf) = 0, so after the softmax every future position gets exactly zero weight and each row still sums to 1 over the allowed ones. A token predicting the next character must not see it — this is a decoder; an encoder would simply leave the mask out.

15.3 0.125

head_size = n_embd // n_head = 384 / 6 = 64, and $64^{-1/2} = 0.125$. Scaling by the head size keeps the scores' variance near 1, so the softmax stays diffuse at initialisation instead of turning nearly one-hot. The gpt.py published with the video multiplied by C**-0.5, where C is the embedding size — $384^{-1/2} \approx 0.051$; a community fix merged on 27 January 2023, ten days after the video, changed it to the head size.